

Show understanding of how data for a bitmapped image are encoded

9608_w21_qp_11_sc_1.b

QUESTION

(b) Identify **three** items that are stored in a **bitmap** file header.

1

.....

2

.....

3

.....

[3]

MARK SCHEME

1(b)	One mark per bullet point to max 3 E.g. • Dimensions e.g.100 × 100 pixels // image size • File size • Colour_depth // bit depth • Location/offset of data within the file • Compression type • Confirmation that it is a bitmap // filetype • Image resolution • Colour palette	3
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9608_s21_qp_12_sc_7.b

QUESTION

(b) The images are stored as bitmap files.

Identify **four** items that will be stored in the header of a bitmap file.

1

2

3

4

[4]

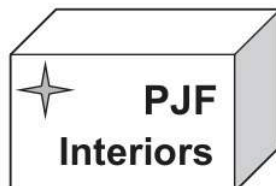
MARK SCHEME

7(b)	1 mark per bullet point to max 4 e.g. • Vertical height in pixels • Horizontal width in pixels • Bit depth/number of bits per pixel • Total file size in bytes • Offset in bytes where the image data is located • Type of compression used	4
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9608_s20_qp_13_sc_7.a

QUESTION

- 7 Xiaoming created the following logo using bitmapped graphics software.



- (a) Describe how **one** typical feature of bitmapped graphics software was used to create the logo.

.....

.....

.....

..... [2]

MARK SCHEME

7(a)	1 mark for each feature and 1 mark for further related expansion to max 2 • Colour select • Select all pixels of the same colour • Add text • To show the name of the company • Resize the star shape • To fit the space available in the box • Fill an area with colour • To shade the side of the box • Select • 'Grab' a number of pixels to perform a task with/to • Copy • Replicate a number of pixels	2
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9608_w19_qp_12_sc_6.d.i

QUESTION

(d) Dominic's tablet captures a video of Dominic to send to other people. The video is made of a sequence of images and a sound file.

(i) Describe how the images and sound are encoded into a digital form.

Images

.....

.....

.....

Sound

.....

.....

.....

[4]

MARK SCHEME

6(d)(i)	1 mark per bullet point to max 4 Max 3 for image Max 3 for sound Images The images are stored as bitmaps Each image is made up of pixels ... each pixel is of a single colour Each colour has a unique binary number Store the sequence of binary numbers for each image / frame // store the binary value of each pixel Sound Measure the height/amplitude of the sound wave A set number of times per second // at regular time intervals Each amplitude has a unique binary number Store the sequence of binary numbers for each sample	4
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9608_w19_qp_11_sc_5.a.i

QUESTION

5 (a) The bit depth of an image dictates how many different colours can be represented by each pixel. (i) State the number of different colours that can be represented by a bit depth of 8 bits. [1]
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MARK SCHEME

5(a)(i)	256	1
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QUESTION

5 Xander creates a presentation that includes images, video and sound.

- (a) The images are bitmap images. A bitmap image can be made up of any number of colours. Each colour is represented by a unique binary number.

Draw **one** line from **each** box on the left, to the correct box on the right to identify the minimum number of bits needed to store each maximum number of colours.

Maximum number of colours

68

256

127

2

249

Minimum number of bits

1

2

3

7

8

9

[3]

MARK SCHEME

5(a)

1 mark for each correctly linked box on the right

Maximum number
of colours

Minimum number
of bits

Diagram showing the relationship between the maximum number of colours and the minimum number of bits required to represent them:

Maximum number of colours	Minimum number of bits
68	7
256	8
127	2
2	1
249	3

3

9608_w18_qp_13_sc_1.a.i

QUESTION

1 A product designer is creating a poster.

(a) The designer creates a 6-colour bitmap image for the poster as shown.

Each colour is represented by a letter, for example, R = red, B = blue.

R	R	P	P	P	G
B	R	R	P	G	G
B	W	B	B	O	O
B	W	W	P	P	O
B	B	R	P	G	O
B	R	R	P	G	O

(i) State the minimum number of bits needed to represent each pixel in the image in **part (a)**.

.....[1]

MARK SCHEME

1(a)(i)	3	1
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QUESTION

- 1 A company is designing a website.
- (a) The company creates a 4-colour bitmap image for the website as shown.
- Each colour is represented by a letter, for example, G = grey, K = black.

G	R	G	K	W	R
G	R	G	K	W	R
G	R	G	K	W	R
G	R	G	K	W	R
G	G	G	K	K	R
W	W	W	W	K	R

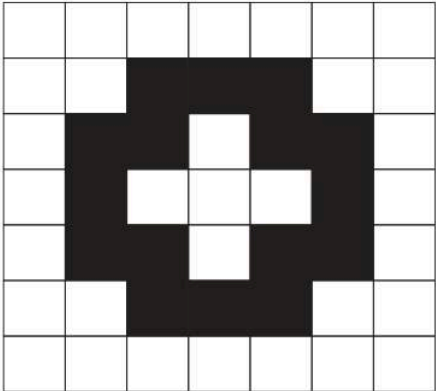
- (i) State the minimum number of bits needed to represent each pixel in the image in **part (a)**.
- [1]

MARK SCHEME

1(a)(i)	2	1
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QUESTION

- 6 A black and white bitmap image is shown.



- (a) State the **minimum** number of bits needed to represent each pixel in this image.
-[1]

MARK SCHEME

6(a)	1	1
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9608_s18_qp_13_sc_6.c

QUESTION

(c) An image has 30 different colours.

State the **minimum** number of bits needed to represent each pixel in the 30-colour image.

.....[1]

MARK SCHEME

6(c)	5	1
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9608_s18_qp_13_sc_6.d

QUESTION

(d) When the image is saved, a header is added to the file.

State the purpose of the **file header**. Give **two** examples of the file header contents.

Purpose

.....

Example 1

.....

Example 2

.....

[3]

MARK SCHEME

6(d)	1 mark for purpose Stores data about the file contents/image/metadata Max 2 marks for examples of contents <u>Confirmation</u> that the file is a BMP // confirmation of file type File size Location / offset of image data within the file Dimensions of the image (in pixels) // <u>image</u> resolution Colour depth (bits per pixel, 1, 4, 8, 16, 24 or 32) Type of compression used (if any)	3
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QUESTION

(e) Graphics software is used to edit a digital photograph.

Give **three** features of graphics software that can be used to edit the photograph.

Describe the effect each has on the photograph.

Feature 1

Effect

.....

.....

Feature 2

Effect

.....

.....

Feature 3

Effect

.....

.....

[6]

MARK SCHEME

6(e)	1 mark for naming tool, 1 mark for describing effect on the photograph e.g. Resize Increase / decrease the size of the image Crop Remove part of the image Blur Reduce the focus Red eye reduction Reduces red (light reflected from human eyes)	6
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QUESTION

2 A logo is designed as a bitmap image.

(a) Describe what is meant by a **bitmap image**.

.....

.....

.....

.....[2]

MARK SCHEME

2(a)	1 mark per bullet, max 2 Made up of pixels Each pixel has one colour Colour of each pixel stored as a binary number	2
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QUESTION

(b) A black and white bitmap image is shown.



(i) Explain how a computer can store this bitmap image.

.....

.....

.....

.....[2]

MARK SCHEME

2(b)(i)	1 mark per bullet, max 2 Each pixel requires only one bit (as there are only two colours) Black represented by 1 and white by 0 (or vice versa) Bits are stored for each pixel in sequence 11111 01010 01010 01010 01010	2
---------	---	----------

9608_s17_qp_13_sc_3.b.i

QUESTION

(b) The computer also has bitmap software.

(i) Define the terms **pixel** and **screen resolution**.

Pixel

.....

Screen resolution

..... [2]

MARK SCHEME

3(b)(i)	<i>Pixel</i> Smallest picture element which can be drawn 1 <i>Screen resolution</i> The number of pixels which can be viewed horizontally and vertically on the screen // or by example - A typical screen resolution is 1680 pixels · 1080 pixels. 1	2
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9608_s17_qp_13_sc_3.b.ii

QUESTION

(ii) A picture has been drawn and is saved as a monochrome bitmap image.

State how many pixels are stored in one byte.

..... [1]

MARK SCHEME

3(b)(ii)	8	1
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9608_s17_qp_13_sc_3.b.iv

QUESTION

- (iv) The actual bitmap file size will be larger than your calculated value.

State another data item that the bitmap file stores in addition to the pixel data.

.....

.....[1]

MARK SCHEME

3(b)(iv)	One from:		1
	Confirmation that the file is a BMP	1	
	File size	1	
	Location/offset of image data within the file	1	
	Dimensions of the image in pixels // image resolution	1	
	Colour depth (bits per pixel)	1	
	Type of compression used, if any	1	

9608_s17_qp_12_sc_3.b.i

QUESTION

- (b) The computer also has bitmap software.

- (i) Define the term **image resolution**.

.....

.....[1]

MARK SCHEME

3(b)(i)	One from:		1
	The <u>number of pixels</u> per <u>unit measurement</u>	1	
	The number of pixels in an image	1	
	The number of pixels wide by the number of pixels high	1	
	Number of pixels per row by the number of rows	1	

9608_s17_qp_12_sc_3.b.ii

QUESTION

- (ii) A picture is drawn and is saved as a 16-colour bitmap image.

State how many bits are used to encode the data for one pixel.

.....[1]

MARK SCHEME

3(b)(ii)	4	1
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9608_s17_qp_12_sc_3.b.iv

QUESTION

- (iv) The actual bitmap file size will be larger than your calculated value as a bitmap file has a file header.

State **two** items of data that are stored in the file header.

1

2[2]

MARK SCHEME

3(b)(iv)	Two from: Confirmation that the file is a BMP File size Location/offset of image data within the file Dimensions of the image (in pixels) // image resolution Colour depth (bits per pixel, 1, 4, 8, 16, 24 or 32) Type of compression used, if any	1 1 1 1 1 1	Max 2
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9608_s17_qp_11_sc_3.b.i

QUESTION

(b) The computer also has bitmap software.

(i) Define the terms **pixel** and **screen resolution**.

Pixel

.....

Screen resolution

.....

[2]

MARK SCHEME

3(b)(i)	<i>Pixel</i> Smallest picture element which can be drawn	1	2
	<i>Screen resolution</i> The number of pixels which can be viewed horizontally and vertically on the screen // or by example - A typical screen resolution is 1680 pixels · 1080 pixels.	1	

9608_s17_qp_11_sc_3.b.ii

QUESTION

(ii) A picture has been drawn and is saved as a monochrome bitmap image.

State how many pixels are stored in one byte.

.....[1]

MARK SCHEME

3(b)(ii)	8	1
----------	---	----------

9608_s17_qp_11_sc_3.b.iv

QUESTION

(iv) The actual bitmap file size will be larger than your calculated value.

State another data item that the bitmap file stores in addition to the pixel data.

.....

.....[1]

MARK SCHEME

3(b)(iv)	One from:		1
	Confirmation that the file is a BMP	1	
	File size	1	
	Location/offset of image data within the file	1	
	Dimensions of the image in pixels // image resolution	1	
	Colour depth (bits per pixel)	1	
	Type of compression used, if any	1	

9608_w16_qp_13_sc_7.b

QUESTION

- (b) The images contained in the magazines are produced using either bitmap or vector graphics software.

Give **four** differences between bitmap and vector graphics.

- 1
-
- 2
-
- 3
-
- 4
-[4]

MARK SCHEME

(b) Four from:

- Bitmap is made up of pixels
// Vector graphic store a set of instructions about how to draw the shape
- Bitmap files are usually bigger than vector graphics files // Take up more memory space
- Enlarging a bitmap can mean the image is pixelated
// vector graphic can be enlarged without the image becoming pixelated
- Bitmap images can be compressed (with significant reduction in file size)
// Vector graphic images do not compress well
- Bitmaps are suitable for photographs / scanned images
// Vector graphics are suitable for more geometric shapes
- Bitmap graphics use less processing power than vector graphics
- Individual elements of a bitmap cannot be grouped
// Individual elements of a vector graphic can be grouped
- Vector graphics need to be 'rasterised' in order to display or print

[4]

9608_w16_qp_11_sc_7.b

QUESTION

(b) The images contained in the magazines are produced using either bitmap or vector graphics software.

Give **four** differences between bitmap and vector graphics.

- 1
- 2
- 3
- 4[4]

MARK SCHEME

(b) Four from:

- Bitmap is made up of pixels
// Vector graphic store a set of instructions about how to draw the shape
- Bitmap files are usually bigger than vector graphics files // Take up more memory space
- Enlarging a bitmap can mean the image is pixelated
// vector graphic can be enlarged without the image becoming pixelated
- Bitmap images can be compressed (with significant reduction in file size)
// Vector graphic images do not compress well
- Bitmaps are suitable for photographs / scanned images
// Vector graphics are suitable for more geometric shapes
- Bitmap graphics use less processing power than vector graphics
- Individual elements of a bitmap cannot be grouped
// Individual elements of a vector graphic can be grouped
- Vector graphics need to be 'rasterised' in order to display or print

[4]

QUESTION

- 8 (a) Six computer graphics terms and seven descriptions are shown below.

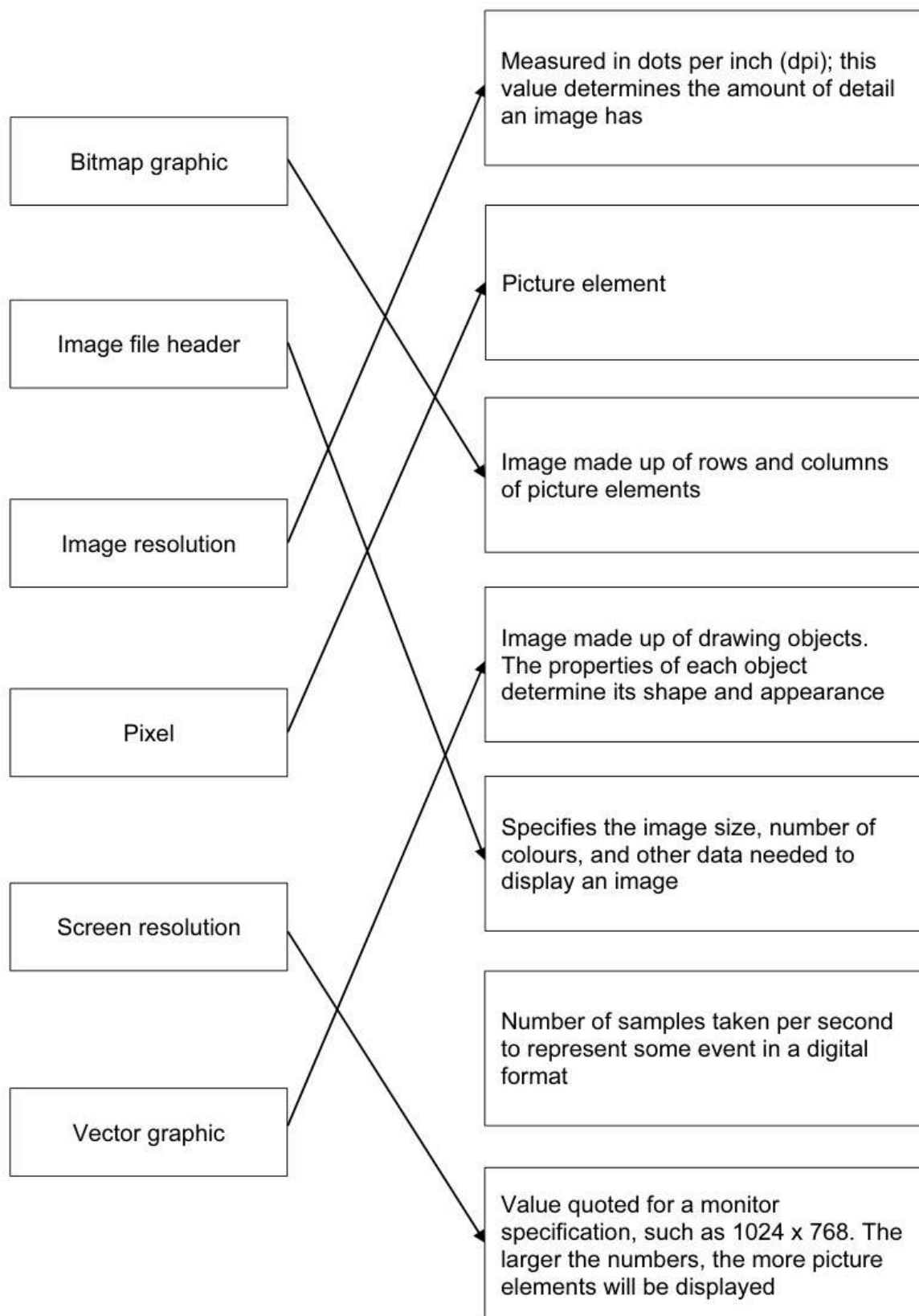
Draw a line to link each term to its correct description.

Term	Description
Bitmap graphic	Measured in dots per inch (dpi); this value determines the amount of detail an image has
Image file header	Picture element
Image resolution	Image made up of rows and columns of picture elements
Pixel	Image made up of drawing objects. The properties of each object determine its shape and appearance.
Screen resolution	Specifies the image size, number of colours, and other data needed to display the image data
Vector graphic	Number of samples taken per second to represent some event in a digital format
	Value quoted for a monitor specification, such as 1024 × 768. The larger the numbers, the more picture elements will be displayed.

[6]

MARK SCHEME

8 (a)



1 mark for each correct line, two lines from one box is incorrect

[6]

9608_w15_qp_11_sc_8.a

QUESTION

- 8 (a) Six computer graphics terms and seven descriptions are shown below.

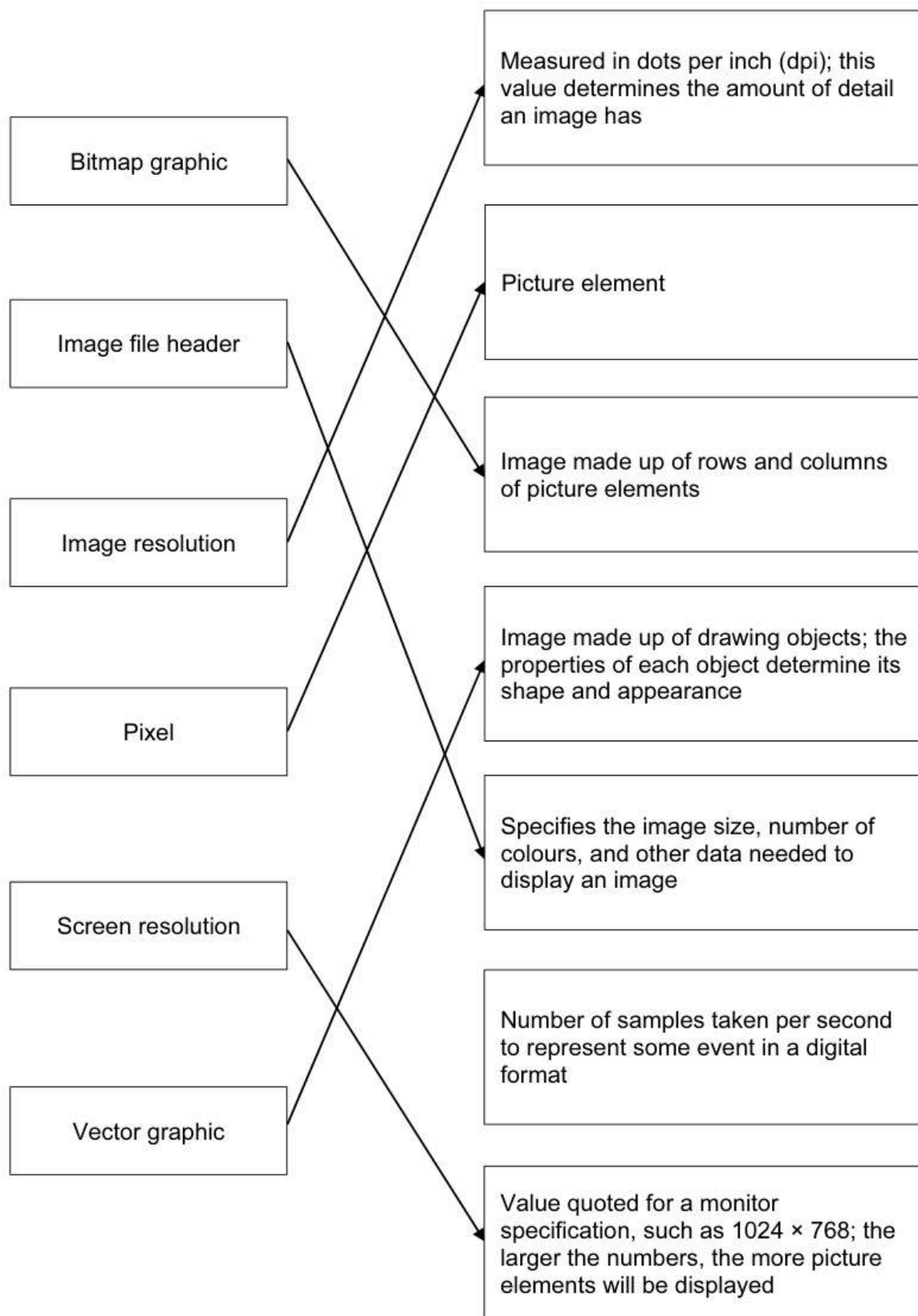
Draw a line to link each term to its correct description.

Term	Description
Bitmap graphic	Measured in dots per inch (dpi); this value determines the amount of detail an image has
Image file header	Picture element
Image resolution	Image made up of rows and columns of picture elements
Pixel	Image made up of drawing objects. The properties of each object determine its shape and appearance.
Screen resolution	Specifies the image size, number of colours, and other data needed to display the image data
Vector graphic	Number of samples taken per second to represent some event in a digital format
	Value quoted for a monitor specification, such as 1024 × 768. The larger the numbers, the more picture elements will be displayed.

[6]

MARK SCHEME

8 (a)



1 mark for each correct line, two lines from one box is incorrect

[6]

512 × 256

Perform calculations to estimate the file size for a bitmap image

9608_s20_qp_13_sc_7.b

QUESTION

(b) The finished logo is 160 pixels wide and 160 pixels high. The image has a colour depth of 3 bytes per pixel.

Calculate an estimate of the file size for the logo. Give your answer in kilobytes. Show your working.

Working
.....
.....

Answer KB

[3]

MARK SCHEME

7(b)	1 mark per bullet point 160 * 160 (= 25600 pixels) 25600 * 3 (= 76800 bytes) 75 KB (divide by 1024) or 76.8 KB (divide by 1000)	3
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QUESTION

- 4 A digital camera takes a bitmap image. The image is 2000 pixels wide by 1000 pixels high with a colour depth of 24-bits.

- (a) Calculate an estimate of the file size for the image. Give your answer in megabytes. Show your working.

Working

.....

.....

.....

.....

Answer MB

[3]

MARK SCHEME

4(a)	1 mark per bullet point • $2000 \times 1000 \times 24 = 48\,000\,000$ bits • $48\,000\,000 / 8 / 1024 / 1024$ • = 6 MB or 5.7 MB	3
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QUESTION

- (ii) Calculate the minimum file size of the image shown in **part (a)**. Show your working.

Working

.....

.....

File size

[3]

MARK SCHEME

1(a)(ii)	1 mark per bullet point • Number of pixels: $6 \times 6 = 36$ • Number of bits: Number of pixels (36) $\times$ 3 • 108 bits / 13.5 bytes	3
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QUESTION

- (b) (i) The designer takes a photograph to put on the poster. The photograph has a resolution of 50 000 pixels by 50 000 pixels. The colours are represented using 4 bytes per pixel.

Estimate the file size of the photograph in gigabytes. Show your working.

Working

.....

.....

.....

Estimated file size

[4]

MARK SCHEME

1(b)(i)	1 mark per bullet point • Number of pixels: $50\,000 \times 50\,000 = 2\,500\,000\,000$ • Number of bytes: Number of pixels $(2\,500\,000\,000) \times \underline{4}$ // $10\,000\,000\,000$ // • Number of bits: Number of pixels $(2\,500\,000\,000) \times \underline{32}$ // $80\,000\,000\,000$ • Conversion to gigabytes • $10\text{ GB} / 9.3\text{ GB}$	4
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QUESTION

- (ii) Calculate the minimum file size of the image shown in **part (a)**. Show your working.

Working

.....

.....

.....

File size

[3]

MARK SCHEME

1(a)(ii)	1 mark per bullet point - Number of pixels: $6 \times 6 // 36$ - Number of bits: Number of pixels (36) $\times 2 \dots$ $\dots = 72$ <u>bits</u> // 9 <u>bytes</u>	3
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9608_w18_qp_12_sc_1.b.i

QUESTION

(b) The company takes a photograph of their office to put on the website. The photograph has a resolution of 1000 pixels by 1000 pixels. Two bytes per pixel are used to represent the colours.

(i) Estimate the file size of the photograph in megabytes. Show your working.

Working

.....

.....

.....

Estimated file size [4]

MARK SCHEME

1(b)(i)	1 mark per bullet point - Number of pixels: $1000 \times 1000 // 1\,000\,000$ - Number of bytes: Number of pixels (1 000 000) $\times 2 // 2\,000\,000 //$ Number of bits: Number of pixels (1 000 000) $\times 16 // 16\,000\,000$ - Conversion to <u>megabytes</u> 2 (MB) // 1.91 (MB)	4
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QUESTION

- (c) The finished logo is 500 pixels by 1000 pixels and uses 35 different colours.

Estimate the file size for the logo. Give your answer in kilobytes. Show your working.

Working

.....

.....

.....

.....

Answer

[4]

MARK SCHEME

2(c)	1 mark per bullet Number of pixels 500*1000 (= 500 000) 35 colours require 6 bits per pixel Number of bytes (500 000 * 6) / 8 = 3 000 000 / 8 (= 375 000) = 375 Kb	4
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QUESTION

- (iii) A second picture has width 2048 pixels and height 512 pixels. It is saved as a 256-colour image.

Calculate the file size in kilobytes.

Show your working.

.....

.....

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.....

.....

.....[3]

MARK SCHEME

3(b)(iii)	<i>Working:</i> Max two from: Number of pixels is $2048 \cdot 512$ 1 One pixel will be stored as one byte 1 Number of kilobytes = $(2048 \cdot 512) / 1024$ 1 <i>Answer:</i> One mark: Number of kilobytes = 1024 KB 1	3
-----------	--	----------

9608_s17_qp_12_sc_3.b.iii

QUESTION

(iii) A second picture has width 8192 pixels and height 256 pixels. It is saved as a 256-colour bitmap.

Calculate the file size in kilobytes.

Show your working.

.....

.....

.....

.....

.....[3]

MARK SCHEME

3(b)(iii)	<i>Working:</i> Max two from: Number of pixels is $8192 \cdot 256$ 1 One pixel will be stored as one byte 1 Number of kilobytes = $(8192 \cdot 256) / 1024$ 1 <i>Answer:</i> One mark: Number of kilobytes = 2048 KB 1	3
-----------	--	----------

QUESTION

(iii) A second picture has width 2048 pixels and height 512 pixels. It is saved as a 256-colour image.

Calculate the file size in kilobytes.

Show your working.

.....

.....

.....

.....

.....

.....[3]

MARK SCHEME

3(b)(iii)	<i>Working:</i> Max two from: Number of pixels is 2048×512 One pixel will be stored as one byte Number of kilobytes = $(2048 \times 512) / 1024$ <i>Answer:</i> One mark: Number of kilobytes = 1024 KB	1 1 1 1	3
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QUESTION

(b) (i) A black and white image is 512 pixels by 256 pixels.

Calculate the file size of this image in kilobytes (KB) (1 KB = 1024 bytes).
Show your working.

.....

.....

.....

.....

.....[2]

MARK SCHEME

$$(b) (i) \frac{512 \times 256}{8 \times 1024} = 16 \text{ KB}$$

1 mark for numerator + 1 mark for denominator

[2]

9608_w15_qp_13_sc_8.b.ii

QUESTION

- (ii) Give a reason why it is important to estimate the file size of an image.

.....
.....
.....[1]

MARK SCHEME

- (ii) so it is possible to estimate how many images can be stored / to decide if it can be sent as an email attachment [1]

9608_w15_qp_11_sc_8.b.i

QUESTION

- (b) (i) A black and white image is 512 pixels by 256 pixels.

Calculate the file size of this image in kilobytes (KB) (1 KB = 1024 bytes).
Show your working.

.....
.....
.....
.....
.....[2]

MARK SCHEME

$$(b) (i) \frac{512 \times 256}{8 \times 1024} = 16 \text{ KB}$$

1 mark for numerator + 1 mark for denominator

[2]

QUESTION

- (ii) Give a reason why it is important to estimate the file size of an image.

.....
.....
.....[1]

MARK SCHEME

- (ii) so it is possible to estimate how many images can be stored / to decide if it can be sent as an email attachment [1]

Show understanding of the effects of changing elements of a bitmap image on the image quality and file size

9608_s20_qp_11_sc_4.b

QUESTION

(b) A second image is taken, this time in black and white. It has the same number of pixels, but the file size is smaller.

Explain why the file size is smaller.

.....

.....

.....

..... [2]

MARK SCHEME

4(b)	1 mark per bullet point to max 2 • Only 1 bit needed to store the colour of each pixel ... • ... so number of pixels * bit depth is $2000 * 1000 * 1$ (rather than $2000 * 1000 * 24$) • ... so the calculation (in part 4(a)) results in smaller figure for file size	2
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QUESTION

(e) Graphics software is used to edit a digital photograph.

Give **three** features of graphics software that can be used to edit the photograph.

Describe the effect each has on the photograph.

Feature 1

Effect

.....

.....

Feature 2

Effect

.....

.....

Feature 3

Effect

.....

.....

[6]

MARK SCHEME

6(e)	1 mark for naming tool, 1 mark for describing effect on the photograph e.g. Resize Increase / decrease the size of the image Crop Remove part of the image Blur Reduce the focus Red eye reduction Reduces red (light reflected from human eyes)	6
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Show understanding of how data for a vector graphic are encoded

9608_w21_qp_11_sc_1.a.i

QUESTION

- 1 An image can be either a bitmap image or a vector graphic.
- (a) Vector graphics are made up of drawing objects and their properties.
- (i) State what is meant by a **drawing object**.

.....
..... [1]

MARK SCHEME

1(a)(i)	An individual shape (in a vector graphic) // a shape defined using mathematics/geometry // a distinct element within an image // possible to alter without affecting the other objects within the image	1
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9608_w21_qp_11_sc_1.a.ii

QUESTION

- (ii) Identify **four** properties of a **drawing object**.

1
.....
2
.....
3
.....
4
..... [4]

MARK SCHEME

1(a)(ii)	One mark per bullet point to max 4 E.g. • Line colour • Line width • Fill colour • Shape • Outline style • Dimensions / size • Position	4
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9608_s21_qp_11_sc_2.a

QUESTION

- 2 Zak designs a logo for his company. He uses vector graphics software to create the logo.



- (a) One of the drawing objects in the logo is a circle.

Identify **four** properties of the circle.

- 1
- 2
- 3
- 4

[4]

MARK SCHEME

2(a)	1 mark per bullet point to max 4 • The position/coordinate of the centre of the circle • Radius • Line width • Line colour • Line style • Fill colour	4
------	---	----------

QUESTION

(b) Describe what is meant by a **drawing list** using the logo as an example.

[2]

MARK SCHEME

2(b)	1 mark per bullet point - A list that stores each separate object in the logo // a list that stores the command/description required to draw each object - Each shape has its own drawing list - Example related to logo, e.g. Three triangles, one square and one circle	2
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QUESTION

(b) Samira uses a computer to draw a logo for her hotel and saves it as a vector graphic. The logo will be placed on the multimedia presentation and elsewhere, such as on signs at the entrance of the hotel.

Samira emails the logo to a company that prints signs, and other documentation for the hotel.

(i) Describe how the logo is represented by the computer.

[3]

MARK SCHEME

1(b)(i)	1 mark per bullet point to max 3 • It is a series of (geometric) shapes / lines / objects • Stored coordinates • Contains a drawing list • Commands / formulae for creating each individual object • and the attributes/property for that object • E.g. the colour, thickness of line etc.	3
---------	--	----------

9608_w16_qp_13_sc_7.b

QUESTION

(b) The images contained in the magazines are produced using either bitmap or vector graphics software.

Give **four** differences between bitmap and vector graphics.

1

.....

2

.....

3

.....

4

.....[4]

MARK SCHEME

(b) Four from:

- Bitmap is made up of pixels
// Vector graphic store a set of instructions about how to draw the shape
- Bitmap files are usually bigger than vector graphics files // Take up more memory space
- Enlarging a bitmap can mean the image is pixelated
// vector graphic can be enlarged without the image becoming pixelated
- Bitmap images can be compressed (with significant reduction in file size)
// Vector graphic images do not compress well
- Bitmaps are suitable for photographs / scanned images
// Vector graphics are suitable for more geometric shapes
- Bitmap graphics use less processing power than vector graphics
- Individual elements of a bitmap cannot be grouped
// Individual elements of a vector graphic can be grouped
- Vector graphics need to be 'rasterised' in order to display or print

[4]

9608_w16_qp_11_sc_7.b

QUESTION

(b) The images contained in the magazines are produced using either bitmap or vector graphics software.

Give **four** differences between bitmap and vector graphics.

- 1
- 2
- 3
- 4[4]

MARK SCHEME

(b) Four from:

- Bitmap is made up of pixels
// Vector graphic store a set of instructions about how to draw the shape
- Bitmap files are usually bigger than vector graphics files // Take up more memory space
- Enlarging a bitmap can mean the image is pixelated
// vector graphic can be enlarged without the image becoming pixelated
- Bitmap images can be compressed (with significant reduction in file size)
// Vector graphic images do not compress well
- Bitmaps are suitable for photographs / scanned images
// Vector graphics are suitable for more geometric shapes
- Bitmap graphics use less processing power than vector graphics
- Individual elements of a bitmap cannot be grouped
// Individual elements of a vector graphic can be grouped
- Vector graphics need to be 'rasterised' in order to display or print

[4]

QUESTION

- 8 (a) Six computer graphics terms and seven descriptions are shown below.

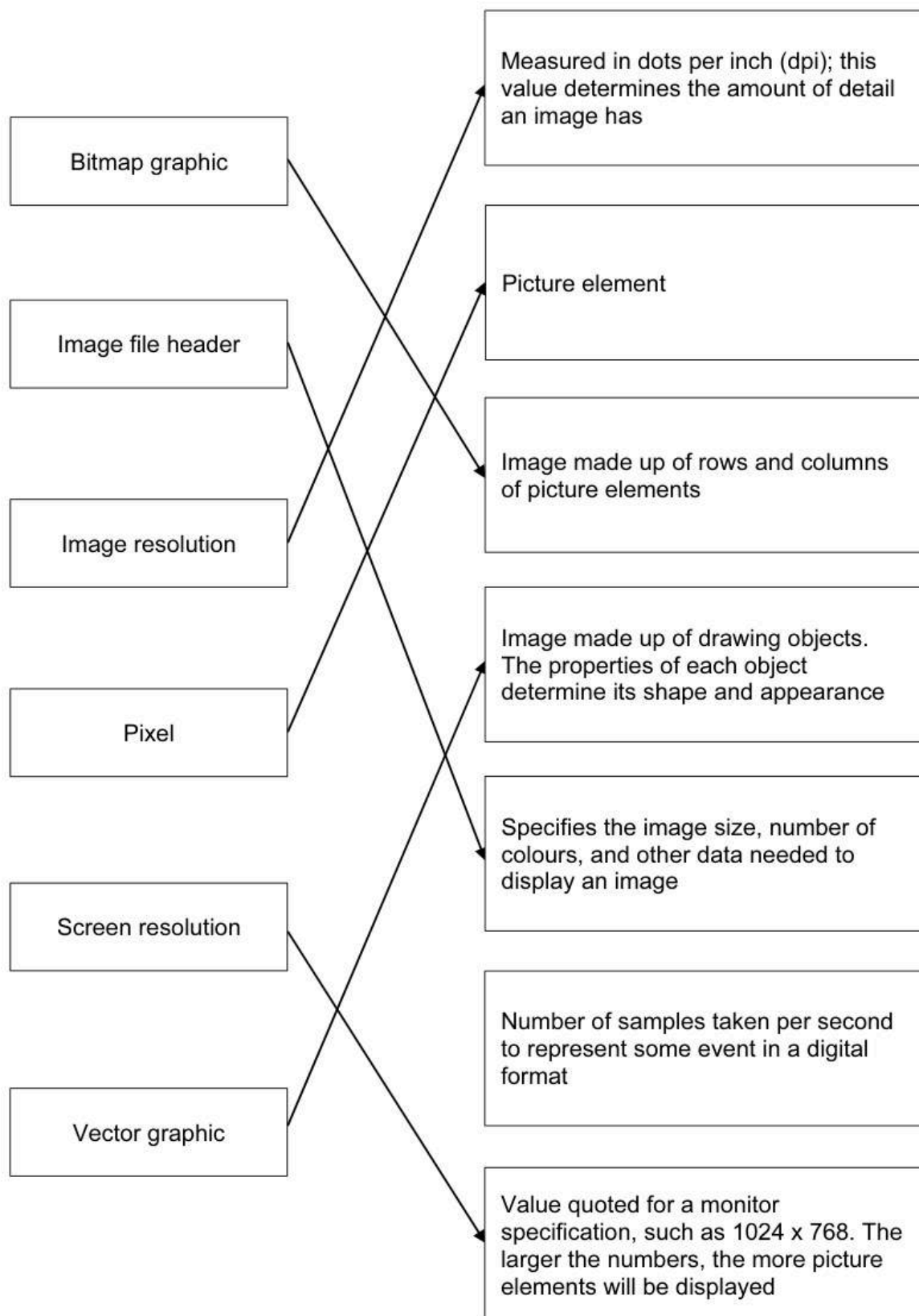
Draw a line to link each term to its correct description.

Term	Description
Bitmap graphic	Measured in dots per inch (dpi); this value determines the amount of detail an image has
Image file header	Picture element
Image resolution	Image made up of rows and columns of picture elements
Pixel	Image made up of drawing objects. The properties of each object determine its shape and appearance.
Screen resolution	Specifies the image size, number of colours, and other data needed to display the image data
Vector graphic	Number of samples taken per second to represent some event in a digital format
	Value quoted for a monitor specification, such as 1024 × 768. The larger the numbers, the more picture elements will be displayed.

[6]

MARK SCHEME

8 (a)



1 mark for each correct line, two lines from one box is incorrect

[6]

9608_w15_qp_11_sc_8.a

QUESTION

- 8 (a) Six computer graphics terms and seven descriptions are shown below.

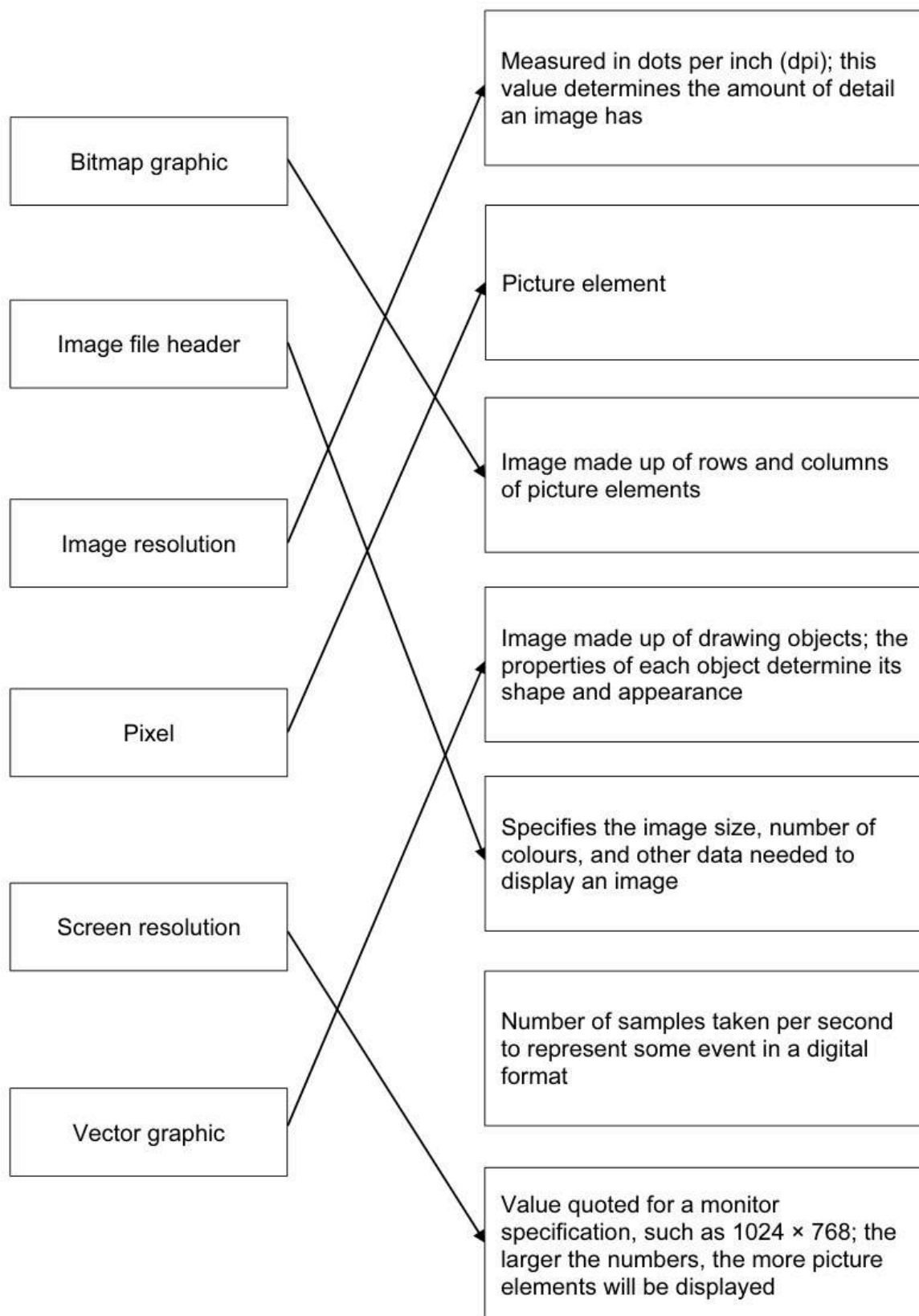
Draw a line to link each term to its correct description.

Term	Description
Bitmap graphic	Measured in dots per inch (dpi); this value determines the amount of detail an image has
Image file header	Picture element
Image resolution	Image made up of rows and columns of picture elements
Pixel	Image made up of drawing objects. The properties of each object determine its shape and appearance.
Screen resolution	Specifies the image size, number of colours, and other data needed to display the image data
Vector graphic	Number of samples taken per second to represent some event in a digital format
	Value quoted for a monitor specification, such as 1024 × 768. The larger the numbers, the more picture elements will be displayed.

[6]

MARK SCHEME

8 (a)



1 mark for each correct line, two lines from one box is incorrect

[6]

512 × 256

Justify the use of a bitmap image or a vector graphic for a given task

9608_s21_qp_11_sc_2.c

QUESTION

(c) Zak could have used a bitmapped image for the logo.

Describe **two** drawbacks of using a bitmapped image for the logo instead of a vector graphic.

Drawback 1

.....

.....

.....

Drawback 2

.....

.....

.....

[4]

MARK SCHEME

2(c)	1 mark per drawback, 1 mark for expansion (max 2 each)	4
	• A bitmap file is likely to take up more storage space • ...because the colour of each pixel needs to be stored • A bitmap image cannot be enlarged // difficult to use in different types of document • ...without the image pixelating • A bitmap would be more difficult to edit • ...because each pixel would need to be edited separately	

QUESTION

- (c) Xiaoming needs to use his logo on his business card, on his website and on large display boards. He is told that he should have created a vector graphic logo instead of a bitmapped graphic logo.

Describe **one** benefit of creating a vector graphic logo instead of a bitmapped graphic logo.

.....

.....

.....

..... [2]

MARK SCHEME

7(c)	1 mark per bullet point to max 2 - The logo can be enlarged without becoming pixelated - because the instructions to create the logo are stored or (Usually) smaller file size - only coordinates and calculations stored instead of individual pixels	2
------	---	----------

QUESTION

- (ii) State **two** reasons why the hotel **logo** is saved as a vector graphic instead of a bitmapped graphic.

1

.....

2

..... [2]

MARK SCHEME

1(b)(ii)	1 mark per bullet point to max 2 - Needs to be large for the signs without becoming pixelated. - Smaller file size means faster transfer rates are possible. - Smaller file size reduces storage requirements when stored many times (on multiple documents).	2
----------	---	----------

QUESTION

(c) The company has created a logo for the website. The logo is a vector graphic.

Describe **two** reasons why a vector graphic is a sensible choice for the logo.

Reason 1

.....

.....

.....

Reason 2

.....

.....

.....

[4]

MARK SCHEME

1(c)	1 mark per bullet point. Max 2 marks for each reason. • Smaller file size • Can be transferred quicker/downloaded quicker • Enlarges without pixilation • Needs to be used on different screens / devices / resolutions	4
------	--	----------

QUESTION

(d) The logo is redesigned as a vector graphic.

State **two** benefits of a vector graphic compared to a bitmap image. Give a reason for each benefit.

Benefit 1

.....

Reason 1

.....

Benefit 2

.....

Reason 2

.....

[4]

MARK SCHEME

2(d)	1 mark per bullet to max 2 marks per benefit Can resize it without pixilation Image is redrawn/recalculated with each adjustment Smaller file size Storing points/equations/commands etc., not individual pixels	4
------	--	----------

Show understanding of how sound is represented and encoded

9608_w21_qp_11_sc_2.a

QUESTION

2 Joanne wants to record sound files and videos for uploading to a social media website.

(a) The following table contains terms about sound representation and encoding.

Complete the table by writing the definitions for each term.

Term	Definition
Sampling	
Sampling resolution	
Sampling rate	

[3]

MARK SCHEME

2(a)	One mark per correct definition	3								
<table><tr><th>Term</th><th>Definition</th></tr><tr><td>Sampling</td><td>Measuring the amplitude of the wave at regular/set time intervals</td></tr><tr><td>Sampling resolution</td><td>The number of bits used to represent each sample</td></tr><tr><td>Sampling rate</td><td>The number of samples taken per unit of time</td></tr></table>			Term	Definition	Sampling	Measuring the amplitude of the wave at regular/set time intervals	Sampling resolution	The number of bits used to represent each sample	Sampling rate	The number of samples taken per unit of time
Term	Definition									
Sampling	Measuring the amplitude of the wave at regular/set time intervals									
Sampling resolution	The number of bits used to represent each sample									
Sampling rate	The number of samples taken per unit of time									

9608_s21_qp_13_sc_9.a.i

QUESTION

9 A website streams music and videos.

(a) Two descriptions about how sound is encoded and represented on a computer are given.

Give the correct term for each description.

(i) "The number of samples taken per unit time"

Term [1]

MARK SCHEME

9(a)(i)	Sampling rate	1
---------	---------------	---

9608_s21_qp_13_sc_9.a.ii

QUESTION

(ii) "The number of bits used to encode each sample"

Term [1]

MARK SCHEME

9(a)(ii)	Sampling resolution // bit depth	1
----------	----------------------------------	---

QUESTION

2 Amir has created a sound file using his desktop computer.

(a) Complete the table by writing the missing definitions and term about sound.

Term	Definition
Sampling	
.....	The number of samples per unit time
Sampling resolution	

[3]

MARK SCHEME

2(a)	1 mark for each correct term/definition	3								
	<table><tr><th>Term</th><th>Definition</th></tr><tr><td>Sampling</td><td>The amplitude of the analogue sound wave is recorded at regular/set intervals</td></tr><tr><td>Sampling rate</td><td>The number of samples per unit time</td></tr><tr><td>Sampling resolution</td><td>The number of bits used to store each sample</td></tr></table>	Term	Definition	Sampling	The amplitude of the analogue sound wave is recorded at regular/set intervals	Sampling rate	The number of samples per unit time	Sampling resolution	The number of bits used to store each sample	
Term	Definition									
Sampling	The amplitude of the analogue sound wave is recorded at regular/set intervals									
Sampling rate	The number of samples per unit time									
Sampling resolution	The number of bits used to store each sample									

QUESTION

(b) Leonardo uses the mobile phone to record his voice.

(i) Describe how sound sampling is used by the mobile phone to encode the sound.

.....

.....

.....

..... [2]

MARK SCHEME

2(b)(i)	1 mark per bullet point to max 2 The amplitude of the wave is measured ... at set, regular time intervals The value is stored as a binary number	2
---------	--	----------

QUESTION

(iii) Leonardo transfers the recordings to his laptop computer. He uses sound editing software to delete some sections of the recordings, and copy and paste to replicate other sections.

Describe **two** other features of sound editing software Leonardo can use to edit the recordings.

1

.....

.....

.....

2

.....

.....

..... [4]

MARK SCHEME

2(b)(iii)	1 mark for naming a feature, 1 mark for description, max 2 marks for each feature e.g. Amplify ... Increase the volume of a section of sound Change pitch ... Increase/decrease frequency of section(s) Change sampling resolution ... to change the accuracy of the sound / file size	4
-----------	--	----------

9608_w19_qp_12_sc_6.d.i

QUESTION

(d) Dominic's tablet captures a video of Dominic to send to other people. The video is made of a sequence of images and a sound file.

(i) Describe how the images and sound are encoded into a digital form.

Images

.....

.....

.....

Sound

.....

.....

.....

[4]

MARK SCHEME

6(d)(i)	1 mark per bullet point to max 4 Max 3 for image Max 3 for sound Images The images are stored as bitmaps Each image is made up of pixels ... each pixel is of a single colour Each colour has a unique binary number Store the sequence of binary numbers for each image / frame // store the binary value of each pixel Sound Measure the height/amplitude of the sound wave A set number of times per second // at regular time intervals Each amplitude has a unique binary number Store the sequence of binary numbers for each sample	4
---------	--	----------

9608_s19_qp_13_sc_5.c

QUESTION

(c) The sound track has a sampling rate of 88.2 kHz and a sampling resolution of 32 bits.

State what is meant by a **sampling rate of 88.2 kHz** and a **sampling resolution of 32 bits**.

Sampling rate of 88.2 kHz

.....

.....

Sampling resolution of 32 bits

.....

.....

[2]

MARK SCHEME

5(c)	1 mark per bullet 88.2 kHz The sound wave is sampled <u>88200</u> times per second 32 bits Each sample is stored as a 32-bit binary number	2
------	--	----------

QUESTION

(d) A sound track is recorded for the video.

(i) Describe how a computer encodes the sound track.

.....

.....

.....

.....

.....

..... [3]

MARK SCHEME

6(d)(i)	1 mark per bullet point to max 3 The amplitude of the wave is measured /sound wave is sampled At <u>set/regular</u> time intervals Each sample is stored as a binary number Samples are stored in order in a file	3
---------	--	----------

QUESTION

(c) The student needs to edit the sound file.

Describe **two** features of sound editing software that can be used to edit the sound file.

Feature 1

.....

.....

.....

Feature 2

.....

.....

..... [4]

MARK SCHEME

1(c)	For 2 features 1 mark for identifying feature, 1 mark for describing what it does. For example: Cut/delete ... Remove part of the sound file Copy and paste ... Replicate part of the sound Amplify ... Increase the volume of a section of sound	4
------	---	----------

9608_s18_qp_12_sc_5.a

QUESTION

5 A student has recorded a sound track for a short film.

(a) Explain how an analogue sound wave is sampled to convert it into digital format.

.....

.....

.....

.....

.....

.....[3]

MARK SCHEME

5(a)	1 mark per bullet to max 3 Amplitude (of the sound wave) measured At <u>set / regular</u> time intervals / per time unit / time period Value of the sample is recorded as a binary number	3
------	---	----------

QUESTION

(d) The student uses sound editing software to edit the sound file.

Name **two** features of sound editing software the student can use to edit the sound file.

Describe the purpose of each feature.

Feature 1

Purpose

.....

.....

Feature 2

Purpose

.....

.....

[4]

MARK SCHEME

5(d)	1 mark for naming feature/tool, 1 mark for description. Max 2 features e.g. Fading Change the volume of a section of the sound for it get louder/quieter Removing sound / elements Delete sections of the sound wave, for example, background noise Copy Repeat elements of the sound wave	4
------	---	----------

QUESTION

3

(a)

A computer has a microphone and captures a voice recording using sound recording software.

Before making a recording, the user can select the sampling rate.

Define the term **sampling rate**. Explain how the sampling rate will influence the accuracy of the digitised sound.

Sampling rate

.....

.....

Explanation

.....

[2]

MARK SCHEME

3(a)	<i>Sampling rate</i> The <u>number of samples</u> taken <u>per unit time</u> // the number of times the amplitude is measured <u>per unit time</u> Increasing the sampling rate will increase the accuracy / precision of the digitised sound // Increasing the sampling rate will result in smaller quantisation errors.	2 1 1
------	---	--

QUESTION

3

(a)

A computer has a microphone and captures a voice recording using sound editing software.

The user can select the sampling resolution before making a recording.

Define the term **sampling resolution**. Explain how the sampling resolution will affect the accuracy of the digitised sound.

Sampling resolution

.....

.....

Explanation

.....

[3]

MARK SCHEME

3(a)	Definition: Max two from: The number of distinct values available to encode/represent each sample 1 Specified by the number of bits used to encode the data for one sample 1 Sometimes referred to as bit depth 1 Explanation: Max two from: A larger sampling resolution will mean there are more values available to store each sample 1 A larger sampling resolution will improve the accuracy of the digitised sound // A larger sampling resolution will decrease the distortion of the sound 1 Increased sampling resolution means a smaller quantization error 1	Max 3
------	---	--------------

9608_s17_qp_11_sc_3.a

QUESTION

3 (a) A computer has a microphone and captures a voice recording using sound recording software.

Before making a recording, the user can select the sampling rate.

Define the term **sampling rate**. Explain how the sampling rate will influence the accuracy of the digitised sound.

Sampling rate

.....

.....

Explanation

.....

[2]

MARK SCHEME

3(a)	Sampling rate The <u>number of samples</u> taken <u>per unit time</u> // the number of times the amplitude is measured <u>per unit time</u> 1 Increasing the sampling rate will increase the accuracy / precision of the digitised sound // Increasing the sampling rate will result in smaller quantisation errors. 1	2
------	--	----------

QUESTION

- 4 A group of students broadcast a school radio station on a website. They record their sound clips (programmes) in advance and email them to the producer.

(a) Describe how sampling is used to record the sound clips.

.....

.....

.....

.....

.....

.....[3]

MARK SCHEME

4 (a) Three from:

[3]

- The height/amplitude of the (sound) wave is determined.
- At set (time) intervals // by example of sensible time period.
- To get an approximation of the sound wave
- And encoded as a sequence of binary numbers // and converted to a digital signal.
- Increasing the sampling rate will improve the accuracy of the recording.

QUESTION

- 4 A group of students broadcast a school radio station on a website. They record their sound clips (programmes) in advance and email them to the producer.

(a) Describe how sampling is used to record the sound clips.

.....

.....

.....

.....

.....

.....[3]

MARK SCHEME

- 4 (a) Three from:** **[3]**
- The height/amplitude of the (sound) wave is determined.
 - At set (time) intervals // by example of sensible time period.
 - To get an approximation of the sound wave
 - And encoded as a sequence of binary numbers // and converted to a digital signal.
 - Increasing the sampling rate will improve the accuracy of the recording.

9608_s15_qp_12_sc_2.a.i

QUESTION

- 2 (a)** Sound can be represented in a computer in a digital format.

- (i)** Give the definition of the term sampling.

.....
.....
.....[1]

MARK SCHEME

- 2 (a) (i)** Any **one** from:
- amplitude of sound wave taken at different points in time
 - measurement of value of analogue signal at regular time intervals/a point in time **[1]**

9608_s15_qp_12_sc_2.a.ii

QUESTION

- (ii)** Give **one** reason why 16-bit sampling is used in an audio compact disc (CD).

.....
.....[1]

MARK SCHEME

(ii) Any **one** from:

- bit depth/sampling resolution sufficient for good quality sound
- higher bit depth/sampling resolution would mean bigger files
- ...hence less (music) content on each CD
- can represent dynamic range of about 90 dB
- 90 dB is basically the maximum dynamic range of human hearing
- compromise between quality and reasonable file size

[1]

9608_s15_qp_12_sc_2.a.iii

QUESTION

(iii) Explain what is meant by the term sampling resolution.

.....

.....

.....

.....[2]

MARK SCHEME

(iii) Any **two** from:

- resolution is the number of distinct values available to encode/represent each sample
- specified by the number of bits used to store/record each sample
- sometimes referred to as bit depth
- the higher the sampling resolution, the smaller the quantization error
- a higher sampling resolution results in less distortion of the sound
- usually 8 bit, 16 bit, 24 bit or 32 bit

[2]

9608_s15_qp_11_sc_2.a.i

QUESTION

2 (a) Sound can be represented in a computer in a digital format.

(i) Give the definition of the term sampling.

.....

.....

.....[1]

MARK SCHEME

2 (a) (i) Any **one** from:

- amplitude of sound wave taken at different points in time
- measurement of value of analogue signal at regular time intervals/a point in time [1]

9608_s15_qp_11_sc_2.a.ii

QUESTION

(ii) Give **one** reason why 16-bit sampling is used in an audio compact disc (CD).

.....
.....[1]

MARK SCHEME

(ii) Any **one** from:

- bit depth/sampling resolution sufficient for good quality sound
- higher bit depth/sampling resolution would mean bigger files
- ...hence less (music) content on each CD
- can represent dynamic range of about 90 dB
- 90 dB is basically the maximum dynamic range of human hearing
- compromise between quality and reasonable file size

[1]

9608_s15_qp_11_sc_2.a.iii

QUESTION

(iii) Explain what is meant by the term sampling resolution.

.....
.....
.....
.....[2]

MARK SCHEME

(iii) Any **two** from:

- resolution is the number of distinct values available to encode/represent each sample
- specified by the number of bits used to store/record each sample
- sometimes referred to as bit depth
- the higher the sampling resolution, the smaller the quantization error
- a higher sampling resolution results in less distortion of the sound
- usually 8 bit, 16 bit, 24 bit or 32 bit

[2]

9608_s15_qp_11_sc_2.b

QUESTION

(b) Describe **two** typical features found in software for editing sound files.

- 1
-
- 2
-[2]

MARK SCHEME

(b) Any **two** from:

- edit start time, stop time and duration of any sound/timeline
- extract/delete/save part of a clip
- frequency, amplitude, pitch alteration
- fade in/out of a clip
- mix/merge multiple sound sources/tracks
- combine different sources at various volume levels
- pan between tracks/channels
- use of filters
- playback to speakers, processors or recording medium
- conversion between different audio file formats
- etc...

[2]

Show understanding of the impact of changing the sampling rate and resolution

9608_w19_qp_13_sc_2.b.ii

QUESTION

- (ii) Leonardo records his voice twice. Each recording is the same length and has the same sampling resolution.

The first recording has a sampling rate of 44 100 Hz. The second recording has a sampling rate of 21 000 Hz.

Describe how the different sampling rates will affect the recording and the sound file.

.....
.....
.....
..... [2]

MARK SCHEME

2(b)(ii)	1 mark per bullet point Sampling resolution of 44100 Hz takes more samples per second, so the file size will be larger // Sampling resolution of 21000 Hz takes fewer samples per second, so the file size will be smaller At a resolution of 44100 Hz, the sound recording is a closer / more accurate representation of Leonardo's voice // At a resolution of 21000 Hz, the sound recording is a less accurate representation of Leonardo's voice	2
----------	---	----------

QUESTION

- (ii) Explain how the sampling rate and sampling resolution affect the file size of the sound track.

Sampling rate

.....

Sampling resolution

.....

[2]

MARK SCHEME

6(d)(ii)	1 mark per bullet point to max 1 for each Sample rate: Increasing the sample rate means more samples per second hence more bits per second <u>and</u> larger file size Decreasing the sample rate means fewer samples per second hence fewer bits per second <u>and</u> smaller file size Sample resolution: A higher sampling resolution means more bits per sample <u>and</u> a larger file size A lower sampling resolution means fewer bits per sample. a smaller file size	2
----------	--	----------

QUESTION

- (b) An analogue-to-digital converter uses sampling to encode the sound.

Explain how different sampling resolutions affect the sound file and the sound it represents.

.....

.....

.....

.....

.....

.....[3]

MARK SCHEME

1(b)	1 mark per bullet point The sampling resolution number of bits used to store each <u>sample</u> Increasing the (sampling) resolution means a larger file size // Decreasing the (sampling) resolution means a smaller file size Increasing the (sampling) resolution gives a more accurate representation of the analogue sound // Decreasing the (sampling) resolution gives a less accurate representation of the analogue sound Increasing the (sampling) resolution means a greater range of values can be stored // Decreasing the (sampling) resolution gives a smaller range of values that can be stored Increasing the (sampling) resolution reduces the quantization errors // Decreasing the (sampling) resolution causes greater quantization errors	3
------	--	----------

9608_s18_qp_12_sc_5.b

QUESTION

(b) Explain the effects of increasing the sampling resolution on the sound file.

.....

.....

.....

.....[2]

MARK SCHEME

5(b)	1 mark per bullet to max 2 (Increasing the sampling resolution means) more bits per sample // larger range of values Larger file size More accurate representation of sound	2
------	--	----------

MARK SCHEME

3(a)	<i>Sampling rate</i> The <u>number of samples</u> taken <u>per unit time</u> // the number of times the amplitude is measured <u>per unit time</u> 1 Increasing the sampling rate will increase the accuracy / precision of the digitised sound // Increasing the sampling rate will result in smaller quantisation errors. 1	2
------	--	---

9608_s17_qp_12_sc_3.a

QUESTION

3 (a) A computer has a microphone and captures a voice recording using sound editing software.

The user can select the sampling resolution before making a recording.

Define the term **sampling resolution**. Explain how the sampling resolution will affect the accuracy of the digitised sound.

Sampling resolution

.....

.....

Explanation

.....

[3]

MARK SCHEME

3(a)	<i>Definition: Max two</i> from: The number of distinct values available to encode/represent each sample 1 Specified by the number of bits used to encode the data for one sample 1 Sometimes referred to as bit depth 1 <i>Explanation: Max two</i> from: A larger sampling resolution will mean there are more values available to store each sample 1 A larger sampling resolution will improve the accuracy of the digitised sound // A larger sampling resolution will decrease the distortion of the sound 1 Increased sampling resolution means a smaller quantization error 1	Max 3
------	---	-------

QUESTION

- 3 (a) A computer has a microphone and captures a voice recording using sound recording software.

Before making a recording, the user can select the sampling rate.

Define the term **sampling rate**. Explain how the sampling rate will influence the accuracy of the digitised sound.

Sampling rate

.....

.....

Explanation

.....

[2]

MARK SCHEME

3(a)	<i>Sampling rate</i> The <u>number of samples</u> taken <u>per unit time</u> // the number of times the amplitude is measured <u>per unit time</u> Increasing the sampling rate will increase the accuracy / precision of the digitised sound // Increasing the sampling rate will result in smaller quantisation errors.	1 1	2
------	--	-------------------	---

QUESTION

- (ii) Give **one** reason why 16-bit sampling is used in an audio compact disc (CD).

.....

.....[1]

MARK SCHEME

- (ii) Any **one** from:

- bit depth/sampling resolution sufficient for good quality sound
- higher bit depth/sampling resolution would mean bigger files
- ...hence less (music) content on each CD
- can represent dynamic range of about 90 dB
- 90 dB is basically the maximum dynamic range of human hearing
- compromise between quality and reasonable file size

[1]

9608_s15_qp_12_sc_2.a.iv

QUESTION

(iv) Give **one** benefit and **one** drawback of using a higher sampling resolution.

Benefit

.....

Drawback

.....[2]

MARK SCHEME

(iv) **1 mark** for benefit and **1 mark** for drawback.

benefit

- allows for larger dynamic ranges
- ...as dynamic range is approximately six times the bit depth
- more accurate representation/crisper sound quality

drawback

- bigger files/occupies more memory/storage
- longer to transmit data/download music
- greater processing power needed

[2]

9608_s15_qp_11_sc_2.a.ii

QUESTION

(ii) Give **one** reason why 16-bit sampling is used in an audio compact disc (CD).

.....

.....[1]

MARK SCHEME

(ii) Any **one** from:

- bit depth/sampling resolution sufficient for good quality sound
- higher bit depth/sampling resolution would mean bigger files
- ...hence less (music) content on each CD
- can represent dynamic range of about 90 dB
- 90 dB is basically the maximum dynamic range of human hearing
- compromise between quality and reasonable file size

[1]

QUESTION

(iv) Give **one** benefit and **one** drawback of using a higher sampling resolution.

Benefit

.....

Drawback

.....[2]

MARK SCHEME

(iv) **1 mark** for benefit and **1 mark** for drawback.

benefit

- allows for larger dynamic ranges
- ...as dynamic range is approximately six times the bit depth
- more accurate representation/crisper sound quality

drawback

- bigger files/occupies more memory/storage
- longer to transmit data/download music
- greater processing power needed

[2]